TA-2: Event-Driven Serverless File Processing Using AWS CloudFormation

Cloud platform	Amazon Web Services (AWS)
Deployment region	Asia Pacific (Sydney), ap-southeast-2
CloudFormation stack	TA2-Serverless-File-Processing
Lambda function	TA2-File-Processor
SNS topic	FileProcessingAlerts
Implementation and testing	26 October 2026

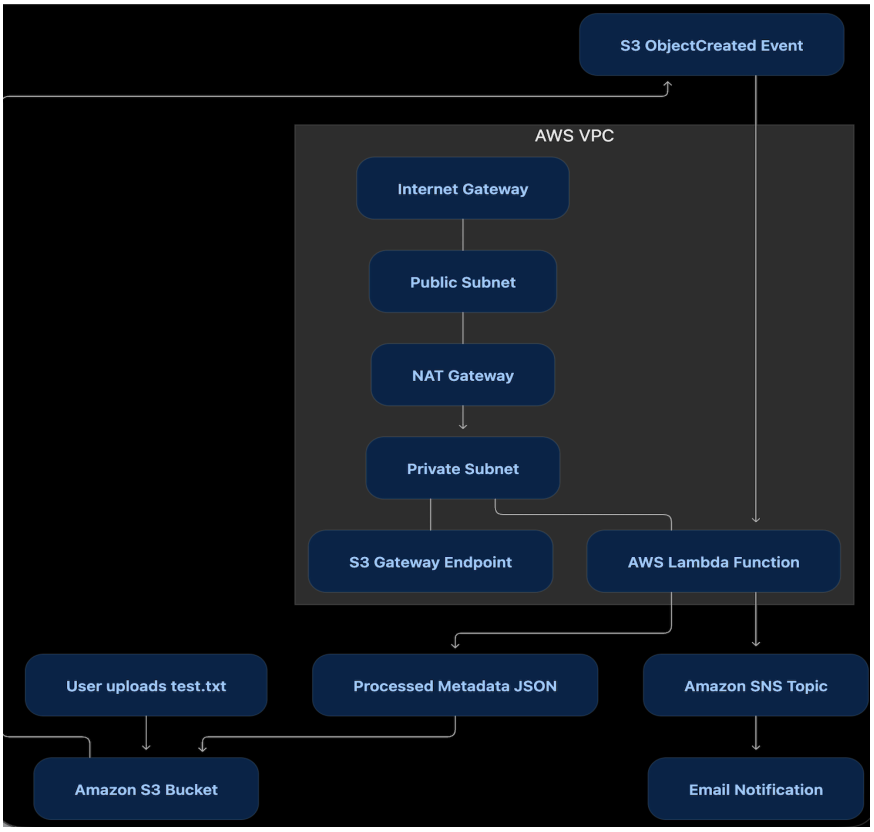
1. Objective

To design and deploy an event-driven serverless file-processing architecture using AWS CloudFormation. An S3 ObjectCreated event triggers a Lambda function running in a private subnet. The function processes the uploaded object's metadata, writes a processing record to S3, and sends a notification using Amazon SNS.

2. AWS Services Used

- AWS CloudFormation
- Amazon S3
- Amazon VPC
- Public and private subnets
- Internet Gateway and NAT Gateway
- S3 Gateway VPC Endpoint
- AWS Lambda
- AWS IAM
- Amazon SNS
- Amazon CloudWatch Logs

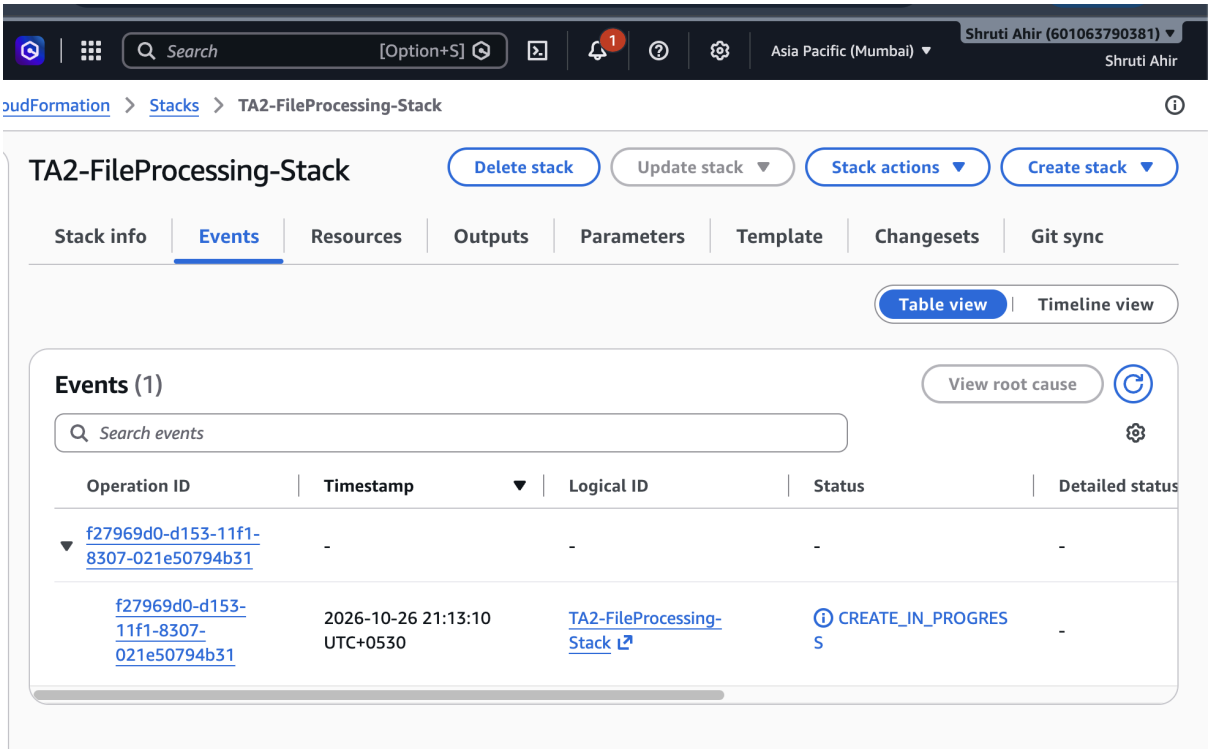
3. Architecture Diagram



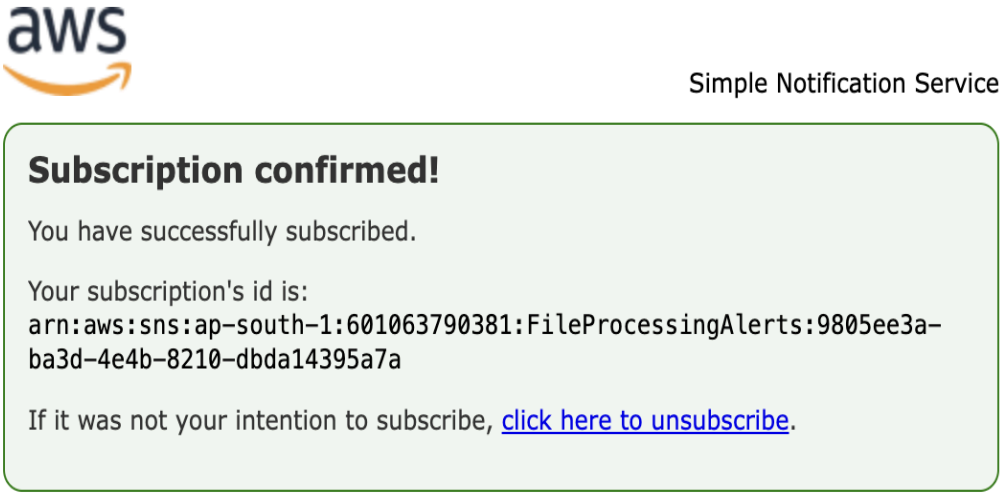
AWS Services and Configuration

Amazon S3	**One private bucket with SSE-S3 (AES256), Block Public Access, and s3:ObjectCreated:* event notifications filtered to uploads/.**
Amazon VPC	VPC 10.0.0.0/16; public subnet 10.0.1.0/24; private subnet 10.0.2.0/24; DNS settings enabled.
Network connectivity	Internet Gateway, public NAT Gateway with Elastic IP, and separate public/private route tables.
AWS Lambda	Python 3.13; 128 MB RAM; 60-second timeout; attached to the private subnet.
AWS IAM	Lambda VPC execution policy; S3 GetObject/PutObject permissions; SNS Publish; S3 invocation permission.
Amazon SNS	FileProcessingAlerts topic with a confirmed email subscription.

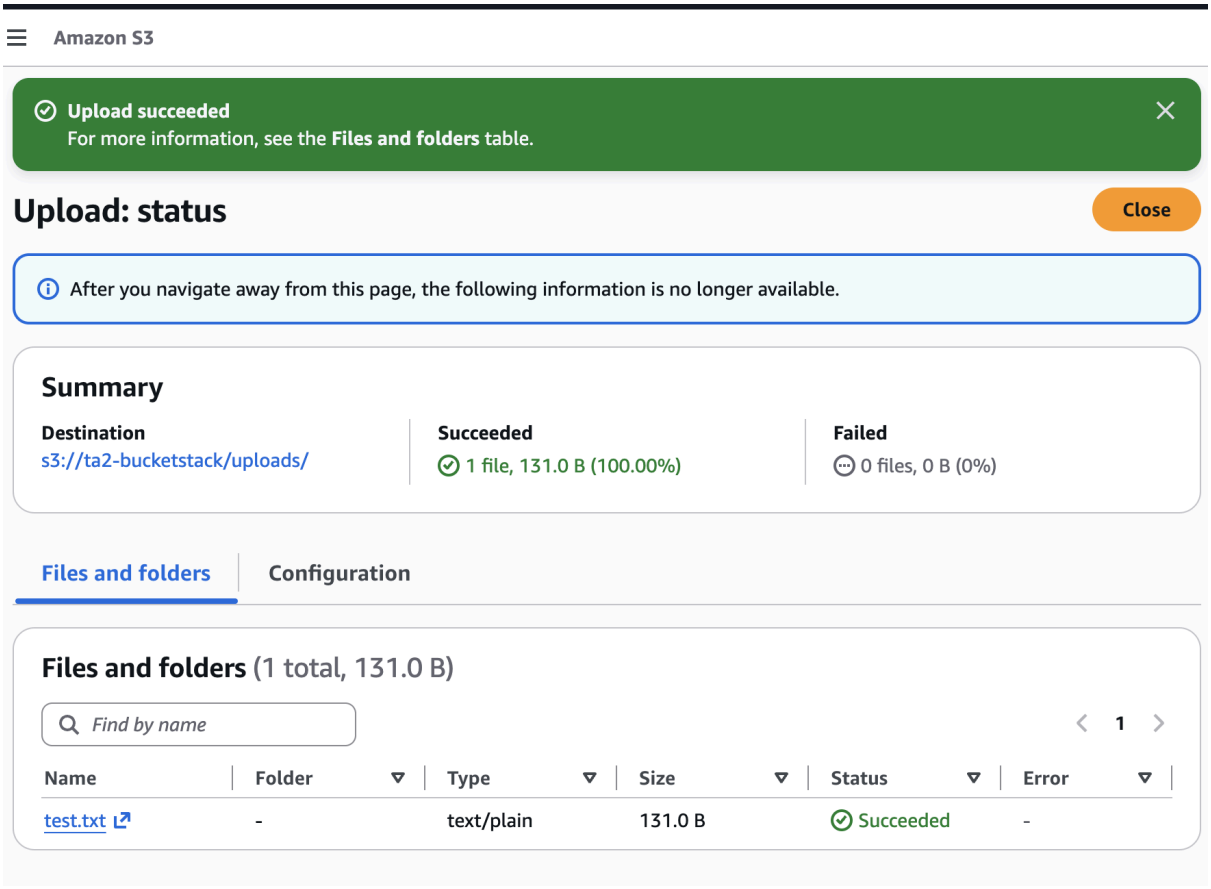
1. CloudFormation Stack Creation Evidence



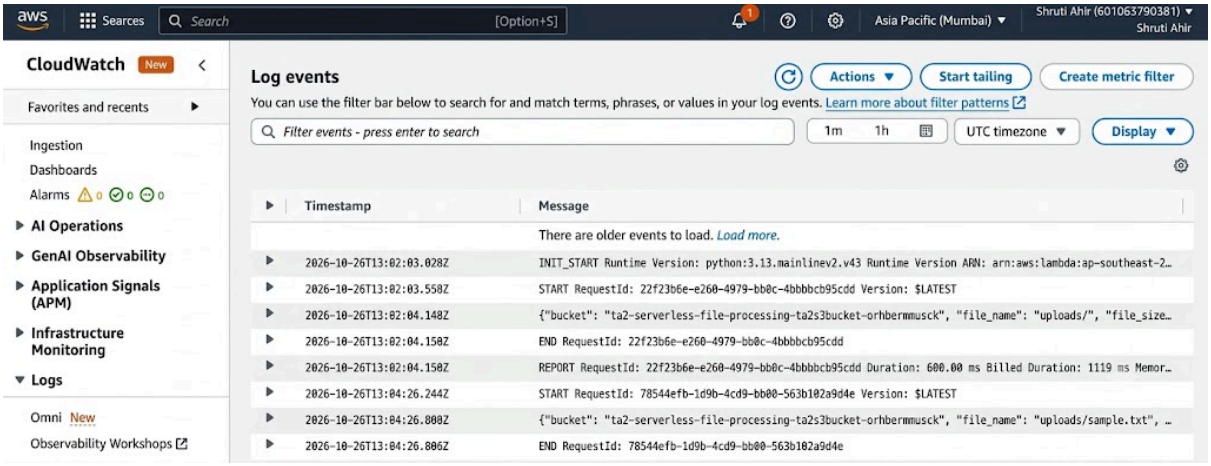
2. SNS Subscription and Stack Update Evidence



3. S3 Upload and Processing Evidence



4. Lambda Execution and SNS Notification





Conclusion

The TA-2 event-driven serverless file-processing system was deployed successfully using AWS CloudFormation. Uploading sample.txt to Amazon S3 invoked the Python Lambda function within the private subnet. The function saved the processed JSON metadata in S3 and sent a successful processing alert through Amazon SNS. CloudFormation status, S3 output, Lambda logs and the received email provide evidence that the required workflow operated successfully.